ONIM: GALE gene, phenotype and variants:

Gene:

- **Gene:** GALE (ONIM 606953)
- **Location:** 1p36.11

Protein:

- **Enzyme**: UDP-galactose-4-epimerase
- **Function:** This gene codifies UDP-galactose-4-epimerase which catalyzes the interconversion of UDP-galactose and UDP-glucose.

Disease:

- **Name:** Galactose epimerase deficiency / Galactosemia III (OMIM 230350)
- **Inheritance:** Autosomal recessive (AR)
- **Symptoms:** sensorineural deafness, hepatomegaly, splenomegaly, vomiting, jaundice, hypotonia, intellectual disability, gross motor developmental delay, language delay, cognitive delay
    - *clarification: These symptoms appear in serious cases, while mild cases hardly show symptoms.

Variants:

Number	Change	Short Notation
.0001	Leu183Pro	**L183P**
.0002	Asn34Ser	**N34S**
.0003	Gly90Glu	**G90E**
.0004	Asp103Gly	**D103G**
.0005	Lys257Arg	**K257R**
.0006	Leu313Met	**L313M**
.0007	Gly319Glu	**G319E**
.0008	Val94Met	**V94M**